Cambridge International AS & A Level Mathematics 9709

S1 Probability & Statistics 1

逐课深度双语学生教材 · 考纲校准版

本册以 Cambridge 9709 Paper 5 的五个官方内容板块为边界。12 课不是考纲规定的固定数目，而是把五个板块按不同认知任务拆成可独立学习的课时；各课知识点和题目数量由内容决定，不设统一配额。

范围与课时审查 | Syllabus mapping

Paper 5 为 1小时15分钟、50分、6–8道结构题。下列12课覆盖官方5.1–5.5；累计频数与编码数据已补入，圆排列、Poisson、抽样估计和假设检验不属于S1。

5.1 Representation of data – 第1–3课：统计量与编码、茎叶/累计频数/箱线图、直方图

5.2 Permutations and combinations – 第4–5课：排列及其限制、组合及其限制

5.3 Probability – 第6–7课：概率法则、条件概率与树图

5.4 Discrete random variables – 第8–10课：一般离散变量、二项分布、几何分布

5.5 The normal distribution – 第11–12课：正态分布、二项分布的正态近似

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第 1 课丨5.1 集中趋势、离散程度与编码数据丨Averages, variation and coded data

深度展开与真题化应用丨In-depth development（共 7 个知识点）

知识点 01/07丨均值、中位数、众数与离散程度丨Measures of location and spread

本课研究数据的中心与离散。mean（均值）使用全部观测值，median（中位数）由排序位置决定，mode（众数）是最常出现的值；range、interquartile range 与 standard deviation 则从不同角度描述 spread。离散或连续、原始或分组的数据会影响可用的计算方法和答案是否只是估计。

English explanation. The mean is a measure of location and the variance is a measure of spread. A frequency is a weight: each value contributes f times to the totals. Always distinguish variance, measured in squared units, from standard deviation, measured in the original unit.

知识点 02/07丨原始、频数与编码数据的公式丨Formulae for raw, frequency and coded data

未分组数据：mean = $\Sigma x/n$。频数表：mean = $\Sigma fx/\Sigma f$。总体方差为 $\mathrm{Var}(X)=\Sigma x^2/n-(\Sigma x/n)^2$；频数表对应 $\Sigma fx^2/\Sigma f-(\Sigma fx/\Sigma f)^2$。standard deviation 是方差的正平方根。若编码 $y=(x-a)/b$，则 $x=a+by$，所以 $\mathrm{mean}(x)=a+b\,\mathrm{mean}(y)$，$\mathrm{Var}(x)=b^2\mathrm{Var}(y)$。

知识点 03/07丨方差快捷公式从何而来丨Deriving the variance identity

方差把每个值与均值的离差平方后取平均。平方有两个作用：正负离差不会抵消，而且离均值更远的值被更重地惩罚。快捷公式由展开 $(x-\mathrm{mean})^2$ 得到，所以两式计算同一个量。

知识点 04/07丨从数据表到均值和标准差丨A reliable calculation sequence

写表格列 x、f、fx、fx^2。先算 Σf 确认总频数；再算 Σfx 得均值；接着算 Σfx^2；最后代入方差公式。若题目给 $\Sigma(x-a)$ 或 $\Sigma(x-a)^2$，先在编码变量上计算，再按 $x=a+by$ 还原。不要把 standard deviation 与 variance 混写：前者与原数据同单位，后者是平方单位。

知识点 05/07丨未分组数据的完整计算丨Worked raw-data calculation

题目丨Question. 数据 3,5,5,7,10 的 mean 和 variance。

正确答案丨Correct answer. $\Sigma x=30$，$n=5$，所以 mean=30/5=6。$\Sigma x^2=9+25+25+49+100=208$。variance$=208/5-6^2=41.6-36=5.6$；sd$=\sqrt{5.6}\approx 2.37$。

知识点 06/07丨用编码总量还原统计量丨Recovering statistics from coded totals

题目丨Question. 20 个数据满足 $\Sigma(x-10)=70$、$\Sigma(x-10)^2=445$。求 mean 与 variance。

正确答案丨Correct answer. 令 $y=x-10$。$\Sigma y=70$，所以 $\mathrm{mean}(y)=70/20=3.5$，$\mathrm{mean}(x)=13.5$。$\mathrm{Var}(x)=\mathrm{Var}(y)=445/20-3.5^2=22.25-12.25=10$；平移10不改变方差。

知识点 07/07丨估计值、单位与标准差口径丨Estimation, units and standard-deviation conventions

计算器的 σx 与 sx 可能分别表示总体和样本标准差。Cambridge 题目要求哪一种取决于题设与考纲口径；手算时必须依照给定公式。分组数据只能用组中值估计，因此答案要写 estimated mean。

逐题精讲丨Exam workshop（共 4 项）

精讲 01/04丨辨认连续数据丨Identifying continuous data

题目丨Question. A delivery company records waiting times of 4.2, 5.7 and 6.1 minutes. State the type of data and explain why.

完整解答 | Full solution. Waiting time is continuous data. Time can, in principle, take any value in an interval; 4.2 minutes is rounded from a measurement rather than obtained by counting. It is therefore not discrete simply because the recorded values have one decimal place.

精讲 02/04 | 由频数表计算方差 | Variance from a frequency table

题目 | Question. The values 2,4,6 have frequencies 1,3,2. Find the variance.

完整解答 | Full solution. $\Sigma f=6$, $\Sigma fx=26$, so mean=13/3. $\Sigma fx^2=4+48+72=124$.
Variance=$124/6-(13/3)^2=35/18\approx 1.94$.

精讲 03/04 | 分组数据为何只能估计 | Why grouped results are estimates

题目 | Question. A student uses class midpoints to calculate a mean of 27.4 and writes “the mean is exactly 27.4”. Correct the conclusion.

完整解答 | Full solution. The value is an estimated mean, not an exact mean. Every observation in a class has been represented by its midpoint, although the original values may lie anywhere within that class. The corrected conclusion is: “The estimated mean is 27.4.”

精讲 04/04 | 线性变换怎样改变统计量 | Effect of a linear transformation

题目 | Question. Every value in a data set is transformed by $y=3x-2$. How do mean and variance change?

完整解答 | Full solution. $E(Y)=3E(X)-2$. $\mathrm{Var}(Y)=3^2\mathrm{Var}(X)=9\mathrm{Var}(X)$; subtracting 2 changes location but not spread.

第 2 课丨5.1 茎叶图、累计频数与箱线图丨Stem-and-leaf, cumulative frequency and box plots

深度展开与真题化应用丨In-depth development（共 6 个知识点）

知识点 01/06丨三种表示分别保留什么信息丨What each representation preserves

ordered stem-and-leaf diagram保留每个原始值；cumulative frequency graph把“不超过某个上组界”的累计人数画成递增曲线；box-and-whisker plot用 minimum、Q1、median、Q3、maximum 概括分布。三种表示承担不同任务，但都要求先明确排序、组界和样本总数。

English explanation. A cumulative frequency curve converts ranks into estimated data values. A box plot then compresses the distribution into five summary values. Compare medians for location and IQRs for consistency, quoting values from the common scale.

知识点 02/06丨从累计频数定位四分位数丨Locating quartiles from cumulative frequency

累计频数逐组相加，横坐标使用 upper class boundary。样本量为 N 时，从累计频数轴读取 N/4、N/2、3N/4 对应的 Q1、median、Q3，并计算 IQR=Q3−Q1。由曲线估计 P(X⩽a) 时，用读出的累计频数除以 N。

知识点 03/06丨画图与读图的完整顺序丨Constructing and reading the diagrams

茎叶图必须按序并写 key，例如 3|7 means 37。累计频数图先算累计列，再以上组界为横坐标作点并平滑连接；读四分位数时从累计频数轴水平到曲线，再垂直到数据轴。箱线图使用同一五数概括，比较两组时必须统一刻度，并分别评论 median 与 IQR。

知识点 04/06丨由有序数据求五数概括丨Five-number summary from ordered data

题目丨Question. 数据 4,7,8,8,10,13,15,19,22，求median、Q1、Q3与IQR。

正确答案丨Correct answer.
n=9，median为第5个值10。下半4,7,8,8的Q1=(7+8)/2=7.5；上半13,15,19,22的Q3=(15+19)/2=17；IQR=9.5。

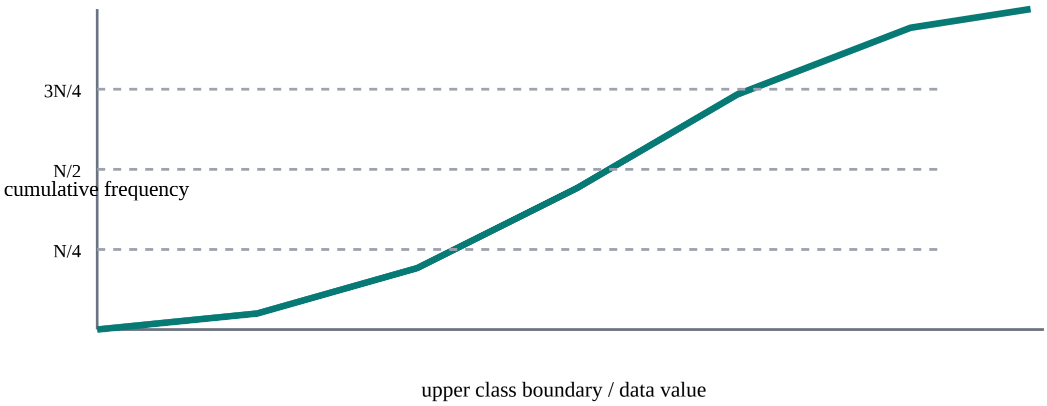
知识点 05/06丨由累计频数读四分位数丨Quartiles from a cumulative-frequency curve

题目丨Question. 80 个数据的累计频数图上，CF=20、40、60 对应 x=18、26、35。求 Q1、median、Q3 与 IQR。

正确答案丨Correct answer. N/4=20、N/2=40、3N/4=60，所以 Q1=18、median=26、Q3=35，IQR=35−18=17。这些数值由曲线估计，精度应与图上刻度一致。

知识点 06/06丨比较分布不能只说“更好”丨Making a supported comparison

“A更好”不是统计结论，除非题目定义数值越大或越小代表更好。应该写 higher/lower median 和 more/less consistent，并引用图中数值。两个箱线图必须使用同一刻度才能视觉比较。



累计频数曲线必须使用上组界；N/4、N/2、3N/4 的水平读数分别定位 Q1、median、Q3。

逐题精讲 | Exam workshop（共 3 项）

精讲 01/03 | 从有序数据求四分位数 | Quartiles from ordered data

题目 | Question. The ordered data are 6, 8, 9, 12, 14, 18, 21, 25. Find the median, Q1, Q3 and IQR.

完整解答 | Full solution. There are 8 values. Median = (12+14)/2 = 13. The lower half is 6,8,9,12, so Q1=(8+9)/2=8.5. The upper half is 14,18,21,25, so Q3=(18+21)/2=19.5. Therefore IQR=19.5−8.5=11.

精讲 02/03 | 用中位数和 IQR 比较两组 | Comparing two distributions

题目 | Question. Explain how to compare two distributions using box plots.

完整解答 | Full solution. Compare medians for location and compare IQRs (or ranges) for spread. Quote both pairs of values and interpret in context; do not infer causation.

精讲 03/03 | 订正没有语境的“更好” | Correcting an unsupported comparison

题目 | Question. A student says “Group A is better because its box is shorter.” Group A has median 72 and IQR 6; Group B has median 65 and IQR 11. Rewrite the comparison accurately.

完整解答 | Full solution. A shorter box shows a smaller IQR, not automatically a “better” group. The supported statement is: Group A has a higher median (72 compared with 65) and is more consistent because its IQR is smaller (6 compared with 11). Whether a higher value is better depends on the context.

第 3 课 | 5.1 直方图与频数密度 | Histograms and frequency density

深度展开与真题化应用 | In-depth development（共 7 个知识点）

知识点 01/07 | 面积代表频数 | Area represents frequency

直方图用于连续分组数据。横轴是 class interval，纵轴通常是 frequency density；柱子的面积而不是高度代表frequency。柱子相接，因为连续变量在组界之间没有类别空隙。

English explanation. In a histogram, area represents frequency. Frequency density corrects for unequal class widths. A taller bar does not necessarily contain more observations; compare areas, not heights, when widths differ.

知识点 02/07 | 频数密度与组距 | Frequency density and class width

frequency density = frequency / class width； 因此 frequency = density × class width。若纵轴给的是其他密度标度，仍以面积成比例为原则。组界必须连续，例如$10<x\leqslant 20$与$20<x\leqslant 35$。

知识点 03/07 | 为什么不能把频数直接当柱高 | Why raw frequency is not the height

当组距不相等时，直接用frequency作高度会让宽组获得过大的面积。除以class width后，每个单位宽度承担相同的频数尺度，于是柱面积 width × density 正好恢复frequency。

知识点 04/07 | 从频数表作图和从图反求频数 | Drawing and reversing a histogram

第一步写每组class width；第二步算density；第三步画连续组界；第四步标纵轴。反向读图时先读高度，再乘组距求frequency。求某个子区间人数时按同一柱内均匀分布作比例估计。

知识点 05/07 | 不等组距的柱高计算 | Unequal-width class calculation

题目 | Question. 区间$0\leqslant t<5$频数20； $5\leqslant t<15$频数30。求两柱高度。

正确答案 | Correct answer. 第一组宽5， density=20/5=4。第二组宽10， density=30/10=3。虽然第二组频数更大，柱高反而较低，因为柱更宽。

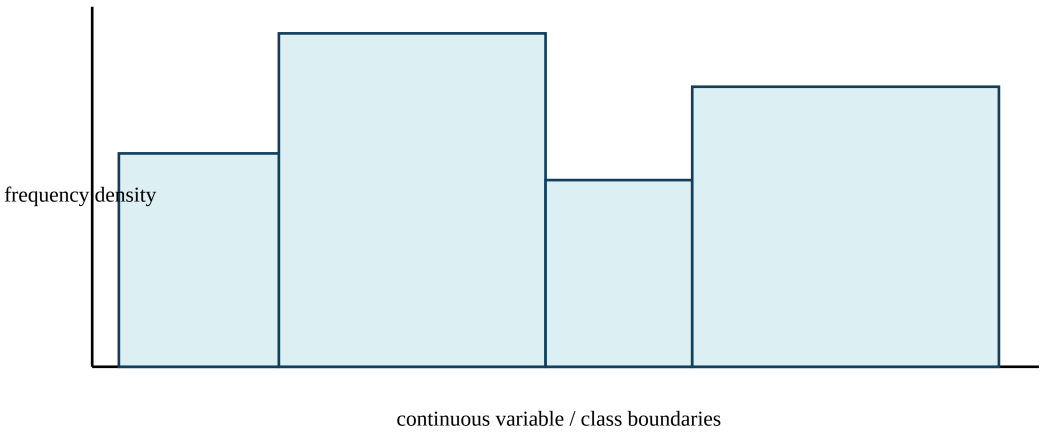
知识点 06/07 | 柱内子区间的频数估计 | Estimating a sub-interval frequency

题目 | Question. $20\leqslant x<30$的柱高2.4； 估计$22\leqslant x<27$的频数。

正确答案 | Correct answer. 整组密度2.4， 子区间宽5， 所以估计频数=2.4 × 5=12。这里使用组内均匀分布假设。

知识点 07/07 | 直方图与条形图的边界 | Histogram versus bar chart

bar chart与histogram不同：bar chart处理类别/离散值，柱间可留空，高度通常表示频数；histogram处理连续分组数据，柱相接，面积表示频数。不要使用组中值画直方图。



柱宽是 class width，柱高是 frequency density，因此每个柱面积恢复该组 frequency。

逐题精讲丨Exam workshop（共 3 项）

精讲 01/03丨不等组距的频数密度丨Frequency density with unequal widths

题目丨Question. A class $10 \leqslant x<18$ has frequency 24 and a class $18 \leqslant x<30$ has frequency 30. Calculate both bar heights.

完整解答丨Full solution. First class width=18−10=8, so density=24/8=3. Second class width=30−18=12, so density=30/12=2.5. The first bar is taller even though its frequency is smaller, because histogram height is frequency per unit width.

精讲 02/03丨由柱高反求频数丨Recovering frequency from a bar

题目丨Question. A histogram bar covers $40 \leqslant x<55$ and has density 3.2. Find its frequency.

完整解答丨Full solution. Class width=15. Frequency=3.2 × 15=48.

精讲 03/03丨订正把频数当柱高的错图丨Correcting histogram heights

题目丨Question. A student draws heights 20 and 30 for classes of widths 5 and 15 because those are their frequencies. Explain and repair the error.

完整解答丨Full solution. Unequal widths make raw frequency an invalid height. Correct densities are 20/5=4 and 30/15=2. Thus the first bar must be twice as tall as the second. Their areas are 5 × 4=20 and 15 × 2=30, which recover the stated frequencies.

第 4 课 | 5.2 排列：顺序为什么重要 | Permutations and factorial notation

深度展开与真题化应用 | In-depth development（共 7 个知识点）

知识点 01/07 | 交换后是否成为新结果 | The order test for permutations

排列（permutation）解决“选出后还要安排顺序”的问题。若从n个不同对象中依次取r个，顺序不同算不同结果，数量为 $nPr=n!/(n-r)!$。$n!=n(n-1)\ldots2\times1$，且$0!=1$。

English explanation. Use a permutation when order creates a different outcome. The formula nPr is a compact form of the product $n(n-1)\ldots$. For repeated objects, divide by the factorial of each repetition because swaps within identical objects have been overcounted.

知识点 02/07 | 阶乘与 nPr 的逐位置含义 | Factorials and the factors in nPr

nPr可展开为 $n(n-1)\ldots(n-r+1)$。例如$8P3=8\times7\times6=336$：第一个位置8种，选定后第二个7种，第三个6种。这个乘法过程就是公式的来源，不是需要死背的符号。

知识点 03/07 | 从乘法原理推出排列公式 | Deriving nPr from the product rule

乘法原理说明：若第一步有a种、每种情况下第二步有b种，则有ab种有序结果。8人选主席、秘书、财务员时三个职位不同，所以A任主席B任秘书与B任主席A任秘书是不同安排。

知识点 04/07 | 限制条件先处理哪个位置 | Handling restricted positions

看到题目先问：是否有位置、次序、排名、座位或密码？若交换两个已选对象会产生新结果，就用排列。然后逐位置写可选数，比直接按 nPr 更能防止限制条件出错。

知识点 05/07 | 不同职位的安排 | Allocating distinct offices

题目 | Question. 8个人中选主席、秘书、财务员，各职位一人，有多少种?

正确答案 | Correct answer.
职位不同，顺序重要。第一职8选1，第二职剩7人，第三职剩6人：$8\times7\times6=336$。也可写 $8P3=8!/(8-3)!=8!/5!=336$。

知识点 06/07 | 重复字母为什么要除以阶乘 | Arrangements with repeated letters

题目 | Question. 排列单词 LEVEL 的字母，有多少种不同排列?

正确答案 | Correct answer. 共有5个位置，若字母都不同有5!。L重复2次、E重复2次，交换相同字母不产生新排列，所以数量$=5!/(2!2!)=120/4=30$。

知识点 07/07 | 排列与组合的分界 | Permutation versus combination

“从8人中选3人”本身没有说明职位，通常是组合；“选主席、秘书、财务员”才是排列。不要看见选人就机械用nCr。重复字母题也不能直接用nPr，因为对象不全不同。

$$\text{slot 1} \times \text{slot 2} \times \text{slot 3}$$

order matters $\rightarrow$ permutation; order does not $\rightarrow$ combination

逐位置相乘显示为什么顺序重要；限制条件应在相应位置减少可选数。

逐题精讲 | Exam workshop（共 3 项）

精讲 01/03 | 从逐位置选择推出 nPr | Deriving nPr from ordered slots

题目 | Question. Eight runners compete for gold, silver and bronze. Calculate the number of possible podium orders from first principles.

完整解答 | Full solution. Gold can be awarded in 8 ways. After that, silver has 7 remaining choices and bronze has 6. By the multiplication principle, total=8 × 7 × 6=336. In compact notation, 8P3=8!/(8−3)!=8!/5!=336.

精讲 02/03 | 相邻限制的捆绑法 | Keeping two objects together

题目 | Question. Six different books are arranged on a shelf. Two particular books must stay together. Find the number.

完整解答 | Full solution. Treat the pair as one block: there are 5 objects to arrange, giving 5!. The pair can be internally ordered in 2! ways. Total=5! × 2=240.

精讲 03/03 | 首位不能为零的密码 | A code with a restricted first digit

题目 | Question. How many 4-digit codes can be made from 0–9 without repetition if the first digit cannot be 0?

完整解答 | Full solution. First digit:9 choices (1–9). Then 9,8,7 choices remain. Total=9 × 9 × 8 × 7=4536. A code restriction is handled position by position.

第 5 课丨5.2 组合与限制条件丨Combinations and restrictions

深度展开与真题化应用丨In-depth development（共 7 个知识点）

知识点 01/07丨只选成员而不安排顺序丨Selections without order

组合（combination）解决“只在乎选了谁、不在乎排列次序”的问题。从n个不同对象选r个，数量nCr=n!/[r!(n−r)!]。分母r!消除同一组选中者的内部排列重复。

English explanation. A combination counts selections, not arrangements. The denominator r! removes the r! orders of the same selected group. For restrictions, translate the wording into disjoint cases or use a complement, then state what each term counts.

知识点 02/07丨nCr 中两个阶乘的含义丨Why the combination formula divides

8C3=8!/(3!5!)。约分：8×7×6×5!/(3×2×1×5!)=(8×7×6)/6=56。答案56表示56个不同委员会，不是56种职位安排。

知识点 03/07丨从排列推导组合丨Deriving nCr from nPr

先按顺序选3人会得到8P3=336，但每个三人委员会被计算3!=6次，因为三人内部有6种顺序。因此8C3=8P3/3!=336/6=56。这就是组合公式中r!的意义。

知识点 04/07丨exactly、at least 与 at most丨Translating restrictions

先做“交换测试”：交换两名已选者，结果是否仍是同一组？若是，用nCr。限制题优先翻译为 exactly、at least、at most，再分互斥情况相加；“至少一个”常用总数减一个都没有。

知识点 05/07丨八人选三名委员丨Choosing an unranked committee

题目丨Question. 8个人中选3名委员，没有职位差别。求方法、公式与答案。

正确答案丨Correct answer.
顺序不重要，用组合。8C3=8!/[3!(8−3)!]=8!/(3!5!)=(8×7×6)/(3×2×1)=56。规范写法：There are 56 possible committees。

知识点 06/07丨至少一个条件的补集法丨Using a complement for at least one

题目丨Question. 5男4女中选3人，至少1女，有多少种？

正确答案丨Correct answer.
总选法9C3=84。减去全男5C3=10，所以至少1女=84−10=74。也可分1女、2女、3女相加，但补集更短。

知识点 07/07丨分情况必须互斥丨Making cases disjoint

at least 1不等于exactly 1；at most
2包含0、1、2。分情况相加前必须保证情况互斥。若先选委员会再分配职位，应先组合再排列，并把两个阶段相乘。

逐题精讲丨Exam workshop（共 2 项）

精讲 01/02丨恰好两名女生的选择丨Selecting exactly two girls

题目丨Question. Choose 4 students from 7 boys and 5 girls, with exactly 2 girls.

完整解答丨Full solution. Choose girls and boys independently: 5C2×7C2=10×21=210. “Exactly 2 girls” forces exactly 2 boys.

精讲 02/02丨先选委员会再选主席丨Selection followed by allocation

题目丨Question. A committee of 4 is chosen from 10 people, then a chair is selected from the committee.

完整解答丨Full solution. Choose committee:10C4. Choose chair from its 4 members:4 choices.
Total=10C4 × 4=840.

第 6 课 | 5.3 概率加法与乘法法则 | Probability rules

深度展开与真题化应用 | In-depth development（共 7 个知识点）

知识点 01/07 | 并集、交集与补集 | Union, intersection and complement

事件$A \cup B$表示A或B至少一个发生；$A \cap B$表示同时发生。互斥事件不能同时发生；独立事件互不改变概率。两组词不能混用。

English explanation. Addition handles an inclusive “or”; multiplication handles a sequence or intersection. Mutually exclusive events have zero intersection. Independent events may occur together, but one event does not change the probability of the other.

知识点 02/07 | 互斥与独立使用不同公式 | Exclusive versus independent events

$P(A \cup B)=P(A)+P(B)-P(A \cap B)$。若互斥，交集为0。独立时$P(A \cap B)=P(A)P(B)$。补集$P(A')=1-P(A)$。

知识点 03/07 | 加法公式为何减去交集 | Avoiding double counting

加法公式减去交集，是因为$P(A)+P(B)$把同时发生部分算了两次。独立乘法来自$P(A|B)=P(A)$：$P(A \cap B)=P(A|B)P(B)=P(A)P(B)$。

知识点 04/07 | 把文字准确翻译成事件 | Translating probability language

先把文字改写成符号：either/or→union，both/and→intersection，not→complement。再判断是否给出mutually exclusive或independent。最后代公式，并检查概率在0到1之间。

知识点 05/07 | 包含交集的并集计算 | Calculating an inclusive union

题目 | Question. $P(A)=0.55$，$P(B)=0.40$，$P(A \cap B)=0.18$，求$P(A \cup B)$。

正确答案 | Correct answer. $0.55+0.40-0.18=0.77$。必须减交集，否则得到0.95并重复计算同时发生部分。

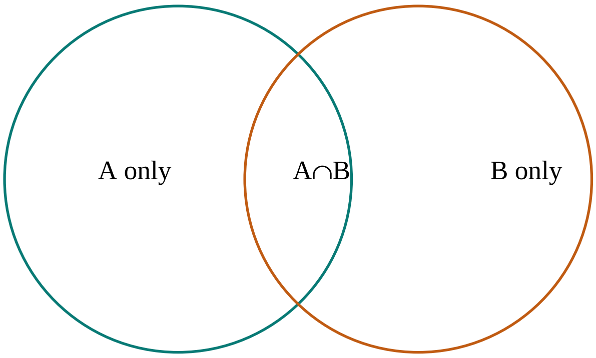
知识点 06/07 | 恰好一个事件发生 | Exactly one of two events

题目 | Question. A、B独立，$P(A)=0.3$，$P(B)=0.6$，求恰好一个发生。

正确答案 | Correct answer.
$P(A \cap B')+P(A' \cap B)=0.3 \times 0.4+0.7 \times 0.6=0.12+0.42=0.54$。两条路径互斥，所以相加。

知识点 07/07 | 非零互斥事件不独立 | Why exclusive events are dependent

互斥且概率都非零的事件不可能独立：若互斥，$P(A \cap B)=0$；若独立，应为$P(A)P(B)>0$。题目中的“or”通常是inclusive or，除非明确说only one。



Venn 图的重叠区是 $A \cap B$；求 $A \cup B$ 时只能计算一次重叠区。

逐题精讲 | Exam workshop（共 3 项）

精讲 01/03 | 含交集的并集概率 | Union with an intersection

题目 | Question. Given $P(A)=0.62$, $P(B)=0.47$ and $P(A \cap B)=0.21$, find $P(A \cup B)$ and explain the subtraction.

完整解答 | Full solution. $P(A \cup B)=P(A)+P(B)-P(A \cap B)=0.62+0.47-0.21=0.88$. The intersection is subtracted once because it was included once in $P(A)$ and once again in $P(B)$.

精讲 02/03 | 检验事件是否独立 | Testing independence

题目 | Question. $P(A)=0.7$, $P(B)=0.5$ and $P(A \cup B)=0.9$. Find $P(A \cap B)$ and decide whether independent.

完整解答 | Full solution. Intersection$=0.7+0.5-0.9=0.3$. Since $P(A)P(B)=0.35 \neq 0.3$, the events are not independent.

精讲 03/03 | 证明互斥不等于独立 | Proving exclusive is not independent

题目 | Question. A student says non-zero mutually exclusive events are independent. Prove that this is false.

完整解答 | Full solution. Mutually exclusive gives $P(A \cap B)=0$. Independence would require $P(A \cap B)=P(A)P(B)$. If both probabilities are non-zero, $P(A)P(B)>0$, contradicting the zero intersection. Therefore non-zero mutually exclusive events are dependent.

第 7 课丨5.3 条件概率与树图丨Conditional probability and tree diagrams

深度展开与真题化应用丨In-depth development（共 7 个知识点）

知识点 01/07丨条件改变样本空间丨A condition changes the sample space

条件概率P(A|B)表示已知B发生后A发生的概率。样本空间被缩小到B，所以分母是P(B)：P(A|B)=P(A∩B)/P(B)。

English explanation. A tree diagram makes conditional structure visible. Multiply along a path, add disjoint paths, and use the total probability of the given event as the denominator when reversing a condition.

知识点 02/07丨树图的分支、路径与总概率丨Branches, paths and total probability

树图每一组分支概率和为1。沿同一路径相乘得到联合概率；满足同一目标的互斥路径相加。不放回抽样会改变第二层分子与分母。

知识点 03/07丨条件概率公式的来源丨Deriving the conditional formula

P(A∩B)既可写P(A|B)P(B)，也可写P(B|A)P(A)。这不是假设独立，而是条件概率定义的重排。只有独立时P(A|B)=P(A)。

知识点 04/07丨不放回时逐层更新概率丨Updating without replacement

第一层写第一次事件；第二层从每个结果分别出发。每条枝先写条件概率，再沿路径乘。题目问given that时，最后用目标交集除以given事件总概率。

知识点 05/07丨两次抽取同色丨Adding two disjoint paths

题目丨Question. 袋中4红3蓝，不放回抽2球，求两球同色。

正确答案丨Correct answer. 红红=4/7 × 3/6=2/7；蓝蓝=3/7 × 2/6=1/7；同色=3/7。两条路径互斥，所以相加。

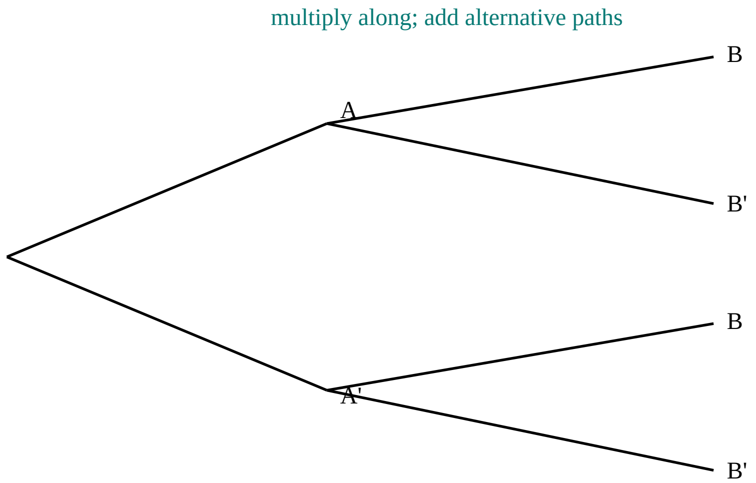
知识点 06/07丨从检测结果反推患病概率丨Reversing a medical-test condition

题目丨Question. 某病患病率0.02；检测对患病者阳性0.95，对未患病者假阳性0.04。求阳性概率。

正确答案丨Correct answer.
P(+)=0.02 × 0.95+0.98 × 0.04=0.019+0.0392=0.0582。若问P(disease|+)，再算0.019/0.0582 ≈ 0.326。

知识点 07/07丨P(A|B) 与 P(B|A) 不可互换丨The direction of conditioning

P(A|B)通常不等于P(B|A)。medical test题中“阳性者患病率”不能直接写检测灵敏度；还必须考虑未患病人口规模和假阳性。



沿一条路径相乘，不同互斥路径相加；第二层概率必须服从第一层已经发生的结果。

逐题精讲 | Exam workshop（共 3 项）

精讲 01/03 | 不放回抽样的条件概率 | Conditional probability without replacement

题目 | Question. A bag contains 5 red and 3 blue balls. Two are drawn without replacement. Find P(second is red | first is blue).

完整解答 | Full solution. Once a blue ball has been removed, 7 balls remain: 5 red and 2 blue. Therefore P(second red | first blue)=5/7. The condition changes the sample space before the second probability is written.

精讲 02/03 | 为什么不能倒置条件 | Why the condition cannot be reversed

题目 | Question. A test has sensitivity 0.9. A student concludes P(disease | positive)=0.9. Explain why this is wrong.

完整解答 | Full solution. Sensitivity is P(positive | disease), not P(disease | positive). Reversing the condition changes the denominator. To find P(disease | positive), calculate the disease-and-positive path and divide by total positive probability, including false positives.

精讲 03/03 | 阳性之后真正患病的概率 | Probability of disease after a positive test

题目 | Question. A disease prevalence is 0.01, sensitivity 0.95 and false-positive rate 0.03. Find and interpret P(disease | positive).

参考答案 | Model answer. $P(D\cap+)=0.01\times0.95=0.0095$. $P(+)=0.0095+0.99\times0.03=0.0392$. Hence $P(D|+)=0.0095/0.0392\approx0.242$. Interpretation: about 24.2% of people who test positive actually have the disease under this model.

第 8 课 | 5.4 离散随机变量与期望方差 | Discrete random variables

深度展开与真题化应用 | In-depth development（共 7 个知识点）

知识点 01/07 | 从随机结果定义 X | Defining a discrete random variable

离散随机变量X把随机结果映射为可数数值。概率分布表必须满足每个$p \geqslant 0$且$\Sigma p=1$。E(X)是长期平均，不要求是X可取的数。

English explanation. A probability distribution is a complete model of a discrete random variable. Expectation describes its long-run average; variance describes spread around that average. A linear change affects location and spread differently.

知识点 02/07 | 概率表、期望与方差公式 | Distribution, expectation and variance

$E(X)=\Sigma xp$；$E(X^2)=\Sigma x^2p$；$Var(X)=E(X^2)-[E(X)]^2$。线性变换$Y=aX+b$时，$E(Y)=aE(X)+b$，$Var(Y)=a^2Var(X)$。

知识点 03/07 | 方差恒等式与线性变换 | Variance identity and transformations

期望是概率加权平均。方差公式由$E[(X-\mu)^2]$展开得到。常数b只是整体平移，不改变离散；乘a把每个离差放大a倍，所以方差放大a^2倍。

知识点 04/07 | 先补概率再计算矩 | A reliable distribution-table method

先用$\Sigma p=1$求未知常数；再增列xp与x^2p；分别求和。若X表示净收益，先从奖金中扣成本，再计算期望，不能把gross prize当profit。

知识点 05/07 | 含未知常数的概率表 | Finding a missing probability

题目 | Question. X取0,1,2，概率k,2k,3k。求k与E(X)。

正确答案 | Correct answer. 6k=1，所以k=1/6。$E(X)=0+1\times2/6+2\times3/6=4/3$。

知识点 06/07 | 从 $E(X^2)$ 得到 Var(X) | Calculating variance from the second moment

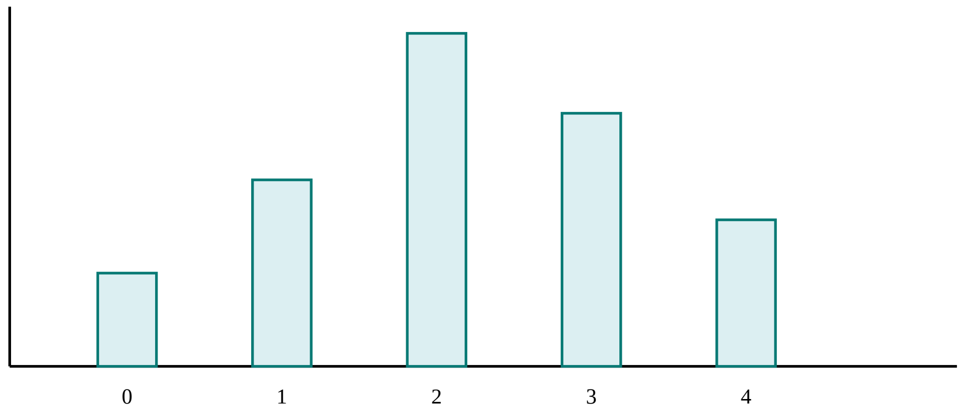
题目 | Question. 上一分布求variance。

正确答案 | Correct answer. $E(X^2)=0+1\times2/6+4\times3/6=7/3$。$Var=7/3-(4/3)^2=5/9$。

知识点 07/07 | 净收益而不是奖金 | Net gain rather than gross prize

$E(X^2)$不等于$[E(X)]^2$。方差出现负数说明算术错误。期望收益为正不保证每次盈利，只说明大量重复后的平均净结果。

probability / frequency representation



柱的位置是 X 的可能取值，柱高是相应概率；全部柱高之和必须为1。

逐题精讲 | Exam workshop（共 3 项）

精讲 01/03 | 补全分布并计算期望方差 | Completing a distribution

题目 | Question. X takes 0, 1, 3 with probabilities k, 2k, 2k. Find k, E(X) and Var(X).

完整解答 | Full solution. Total probability gives k+2k+2k=5k=1, so k=0.2. E(X)=0(0.2)+1(0.4)+3(0.4)=1.6. $E(X^2)$=0+1(0.4)+9(0.4)=4.0. Hence Var(X)=4.0−1.6^2=1.44.

精讲 02/03 | 将奖金改写为净收益 | Modelling net gain

题目 | Question. A game costs $4 and pays $12 with probability 0.25, otherwise $0. A student uses outcomes 12 and 0. Repair the model.

完整解答 | Full solution. The random variable must be net gain, so the outcomes are 12−4=$8 and 0−4=−$4. Expected net gain=0.25(8)+0.75(−4)=2−3=−$1. Using gross prizes would ignore the certain entry cost.

精讲 03/03 | 解释负期望的现实含义 | Interpreting a negative expectation

题目 | Question. Explain an expected net gain of −$1 in English.

参考答案 | Model answer. The result does not mean that the player loses exactly $1 in every game. It means that over a very large number of plays under the same probabilities, the average net gain per game is expected to approach −$1.

第 9 课丨5.4 二项分布：建模、概率与累积事件丨Binomial distribution

深度展开与真题化应用丨In-depth development（共 7 个知识点）

知识点 01/07丨二项模型的四个条件丨Four binomial conditions

X~B(n,p)表示n次Bernoulli试验中的成功次数。必须满足：固定n、每次只有success/failure、试验独立、成功概率p恒定。

English explanation. A binomial variable counts successes. The combination chooses the positions of the successes; the powers give the probability of each success–failure pattern. Always state the random variable and verify the four assumptions.

知识点 02/07丨单点概率、均值与方差丨Probability, mean and variance

P(X=r)=nCr p^r(1−p)^(n−r)。E(X)=np，Var(X)=np(1−p)。符号翻译：at least r是X⩾r；more than r是X>r；at most r是X⩽r。

知识点 03/07丨组合数为何出现在公式中丨Why nCr appears

某一条恰有r次成功的序列概率是p^r(1−p)^(n−r)。成功可以出现在n个位置中的任意r个，共nCr种，因此乘上组合数。

知识点 04/07丨把至少、至多翻译成事件丨Translating cumulative events

第一行定义X并写分布。第二行把文字翻译成事件。第三行决定单项、累积或补集。第四行代公式/计算器。最后写概率并保留3 significant figures左右。

知识点 05/07丨恰好三次成功丨Exactly three successes

题目丨Question. 硬币正面概率0.4，抛8次，恰3次正面。

正确答案丨Correct answer. 令X为正面次数，X~B(8,0.4)。P(X=3)=8C3(0.4)^3(0.6)^5≈0.2787，答案0.279。

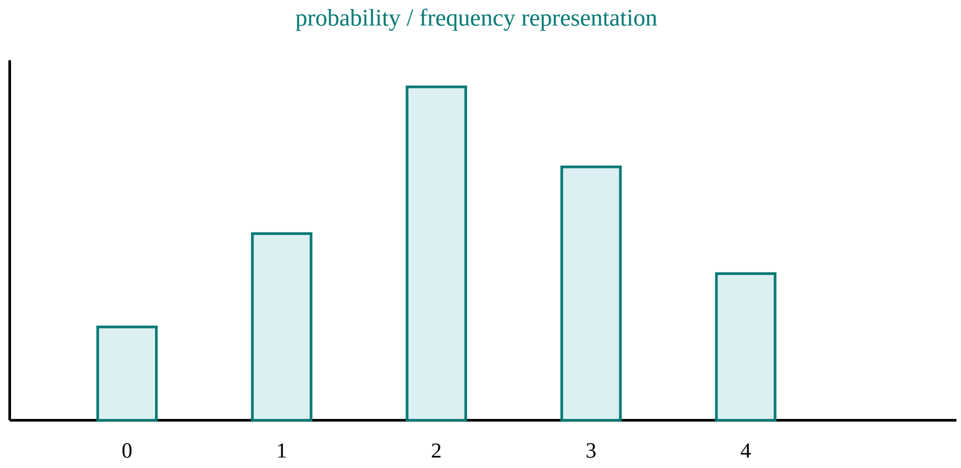
知识点 06/07丨至少一次的补集法丨Complement for at least one

题目丨Question. 同一模型，至少1次正面。

正确答案丨Correct answer. P(X⩾1)=1−P(X=0)=1−(0.6)^8≈0.9832。用补集只需一项。

知识点 07/07丨不放回通常破坏二项条件丨Why sampling without replacement is different

“不放回抽样”通常使p改变，不是严格二项。at least 3不能输入P(X=3)。计算器binomcdf常给P(X⩽r)，求P(X⩾r)需1−P(X⩽r−1)。



二项变量取 0 到 n 的整数值；柱形展示离散概率而不是连续面积。

逐题精讲丨Exam workshop（共 3 项）

精讲 01/03丨解释二项概率中的每个因子丨Explaining every binomial factor

题目丨Question. A biased coin has P(head)=0.3 and is tossed 6 times. Find P(exactly 2 heads) by explaining every factor.

完整解答丨Full solution. Choose the 2 head positions in 6C2 ways. Any fixed pattern with 2 heads and 4 tails has probability $0.3^2 \times 0.7^4$. Hence $P(X=2)=6C2(0.3)^2(0.7)^4=15 \times 0.09 \times 0.2401 \approx 0.324$.

精讲 02/03丨订正 at least 的计算器输入丨Correcting an at-least event

题目丨Question. A student enters binomcdf(n,p,3) for “at least 3 successes”. Correct the event and input logic.

完整解答丨Full solution. binomcdf(n,p,3) gives $P(X \leqslant 3)$, which is the wrong tail. “At least 3” means $X \geqslant 3$. Use the complement $P(X \geqslant 3)=1-P(X \leqslant 2)$, so enter 1−binomcdf(n,p,2).

精讲 03/03丨判断不放回抽样能否建模丨Checking the binomial conditions

题目丨Question. State whether sampling 10 items without replacement from a batch of 20 gives a binomial model.

参考答案丨Model answer. It is not an exact binomial model because removing an item changes the composition of the batch and therefore changes the success probability on later selections. The trials are also not independent.

第 10 课丨5.4 几何分布与等待时间丨Geometric distribution

深度展开与真题化应用丨In-depth development（共 6 个知识点）

知识点 01/06丨第一次成功的等待模型丨Waiting for the first success

几何分布描述独立重复试验中第一次成功出现的试验编号。若每次成功概率p不变，X的可能值为1,2,3,...。

English explanation. A geometric variable is a waiting-time model. Count the failures before the first success carefully. “On trial r” includes a final factor p, while “after trial r” means r failures and does not.

知识点 02/06丨on、after 与 within 的公式丨Formulae for three common wordings

P(X=r)=(1-p)^(r-1)p; P(X>r)=(1-p)^r; P(X⩽r)=1-(1-p)^r。E(X)=1/p。

知识点 03/06丨r-1 次失败后再成功丨Deriving the geometric probability

第一次成功在第r次意味着前r-1次全部失败，随后成功，所以概率是(1-p)^(r-1)p。超过r次仍未成功意味着前r次全失败。

知识点 04/06丨先数失败次数丨Counting failures before success

先分清“on trial r”“after trial r”“within r trials”。写出失败次数，再写最后是否需要乘p。检查X从1开始，不存在第0次首次成功。

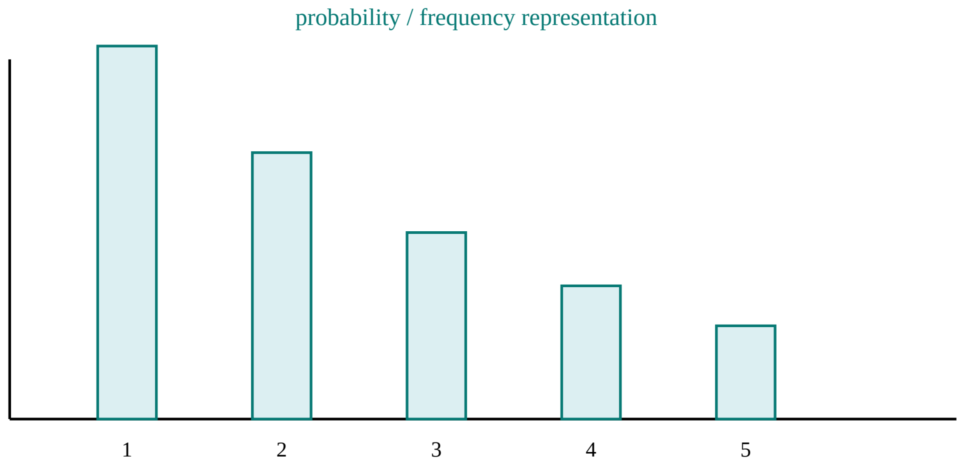
知识点 05/06丨首次成功在第4次丨First success on trial four

题目丨Question. 每次成功率0.25，首次成功在第4次。

正确答案丨Correct answer. 前三次失败、第四次成功：P(X=4)=0.75^3 × 0.25 ≈ 0.1055。

知识点 06/06丨成功概率必须保持不变丨The constant-probability condition

P(X=4)不是0.75^4，也不是0.25^4。前者表示前4次全失败，后者表示4次全成功。题目若成功概率随次数改变，就不能使用几何模型。



几何分布从1开始，随着首次成功所需试验次数增加，概率按固定比例下降。

逐题精讲丨Exam workshop（共 3 项）

精讲 01/03丨首次成功在第5次丨First success on trial five

题目丨Question. Success probability is 0.2. Find the probability that the first success occurs on trial 5.

完整解答丨Full solution. Trials 14 must fail and trial 5 must succeed. Thus P(X=5)=0.8^4 × 0.2=0.08192. There are four failure factors, not five, because the fifth trial is the success.

精讲 02/03丨订正失败次数丨Correcting the number of failures

题目丨Question. A student writes $P(X=4)=0.7^4$ when $p=0.3$. Explain exactly which event 0.7^4 represents and repair the answer.

完整解答丨Full solution. 0.7^4 means the first four trials all fail, so it represents $P(X>4)$. For the first success on trial 4, only the first three trials fail and the fourth succeeds: $P(X=4)=0.7^3\times 0.3=0.1029$.

精讲 03/03丨区分 on 与 after丨Distinguishing on from after

题目丨Question. Explain the difference between "on the sixth trial" and "after the sixth trial".

参考答案丨Model answer. "On the sixth trial" requires five failures followed by a success, so its probability is $(1-p)^5p$. "After the sixth trial" means no success in the first six trials, so its probability is $(1-p)^6$.

第 11 课 | 5.5 正态分布：标准化与反求界值 | Normal distribution

深度展开与真题化应用 | In-depth development（共 7 个知识点）

知识点 01/07 | 均值、方差与曲线面积 | Mean, variance and area

$X\sim N(\mu,\sigma^2)$是连续、对称、钟形分布。μ决定中心，σ决定伸展程度；第二参数σ^2是variance。曲线总面积1，单点概率为0。

English explanation. A normal probability is an area. A sketch identifies the correct tail; standardisation measures distance from the mean in standard deviations. In inverse problems, use the area to determine the sign of z before converting back.

知识点 02/07 | 标准化与反标准化 | Standardising and converting back

标准化$Z=(X-\mu)/\sigma$，把任意正态变量转换为$N(0,1)$。$P(X<a)=P[Z<(a-\mu)/\sigma]$。反求a时先从概率得到z，再用$a=\mu+z\sigma$。

知识点 03/07 | 为什么除以标准差 | Why standardisation divides by sigma

减μ把中心移到0；除σ把一个标准差重新缩放为1。标准化不改变某点处于均值左侧还是右侧，因此z的正负必须与草图一致。

知识点 04/07 | 先画阴影再判断尾部 | Sketching before calculation

画钟形草图，标μ、界值与阴影；写分布；标准化；读表/计算器；利用对称或补集；最后检查概率大小。反求界值时先判断z应正还是负。

知识点 05/07 | 已知界值求概率 | Finding an area from a boundary

题目 | Question. $X\sim N(50,3^2)$，求$P(X<56)$。

正确答案 | Correct answer.
$z=(56-50)/3=2$。$P(X<56)=P(Z<2)=0.9772$。56在均值右侧2个sd，左侧概率应大于0.5，检查通过。

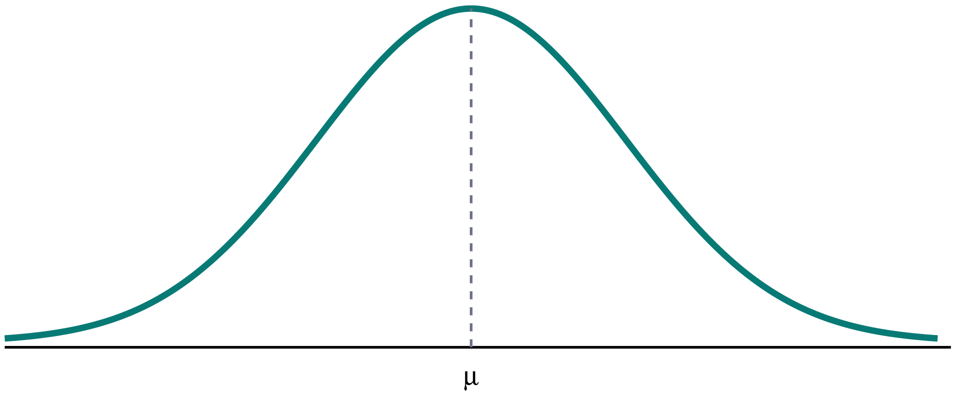
知识点 06/07 | 已知概率反求界值 | Finding a boundary from an area

题目 | Question. 同一分布，求a使$P(X<a)=0.10$。

正确答案 | Correct answer.
下尾10%对应$z\approx-1.2816$。$a=50+(-1.2816)\times3\approx46.2$。下尾界值应小于均值，检查通过。

知识点 07/07 | 第二参数是方差 | The second parameter is variance

$N(50,9)$表示variance=9、sd=3，不是sd=9。upper tail
0.1对应z正值，下尾0.1对应负值。不要把density高度当probability。



正态概率对应曲线下的面积；均值决定中心，标准差决定曲线伸展。

逐题精讲 | Exam workshop（共 3 项）

精讲 01/03 | 由草图检查尾部方向 | Checking the tail with a sketch

题目 | Question. $X\sim N(70,8^2)$. Find $P(X>82)$ with a sketch-based sign check.

完整解答 | Full solution. 82 is above the mean, so the required upper tail must be below 0.5. Standardise: $z=(82-70)/8=1.5$. Thus $P(X>82)=P(Z>1.5)=1-\Phi(1.5)=1-0.9332=0.0668$.

精讲 02/03 | 订正方差与标准差混用 | Correcting variance and standard deviation

题目 | Question. For $X\sim N(40,25)$, a student uses $\sigma=25$. Correct the calculation of $P(X<45)$.

完整解答 | Full solution. The second parameter is variance, so $\sigma=\sqrt{25}=5$. Then $z=(45-40)/5=1$ and $P(X<45)=\Phi(1)=0.8413$. Using 25 confuses variance with standard deviation and gives the wrong scale.

精讲 03/03 | 由上尾概率反求界值 | Inverse normal from an upper tail

题目 | Question. Find a such that $P(X>a)=0.10$ for $X\sim N(100,12^2)$, and explain the sign of z.

参考答案 | Model answer. An upper-tail probability of 0.10 means a lies above the mean, so z is positive. $\Phi(z)=0.90$ gives $z\approx1.2816$. Therefore $a=100+1.2816(12)\approx115.4$.

第 12 课 | 5.5 二项分布的正态近似 | Normal approximation to binomial

深度展开与真题化应用 | In-depth development（共 7 个知识点）

知识点 01/07 | 何时可用正态近似 | Conditions for the approximation

当n较大且p不过分接近0或1时，B(n,p)的柱状分布可用N(np,np(1−p))近似。常用检查是np和n(1−p)都足够大。

English explanation. Continuity correction aligns whole discrete bars with a continuous area. Write the original discrete event, the corrected continuous event and the approximating normal distribution so the logic is visible.

知识点 02/07 | 近似分布的均值与方差 | Parameters of the approximating normal

连续性修正把离散整数边界移0.5：$P(X\leqslant 70)\to P(Y<70.5)$；$P(X\geqslant 70)\to P(Y>69.5)$；$P(60\leqslant X\leqslant 70)\to P(59.5<Y<70.5)$。

知识点 03/07 | 连续性修正对应完整柱形 | Why continuity correction uses half-units

离散值70可想成覆盖69.5到70.5的柱。若连续曲线要包含整个“70”柱，边界必须放在70.5；这就是continuity correction，不是随意加减。

知识点 04/07 | 从离散事件到连续区间 | Translating the event before standardising

定义$X\sim B(n,p)$，检查条件；设$Y\sim N(np,np(1-p))$；翻译事件并做±0.5修正；标准化两个界值；算面积；写approximately。

知识点 05/07 | 单尾概率的完整近似 | A complete one-tail approximation

题目 | Question. $X\sim B(200,0.4)$，近似求$P(X\leqslant 70)$。

正确答案 | Correct answer.
$np=80$，$nq=120$，条件合适。$Y\sim N(80,48)$。$P(X\leqslant 70)\approx P(Y<70.5)$。$z=(70.5-80)/\sqrt{48}\approx -1.371$，概率≈0.0851。

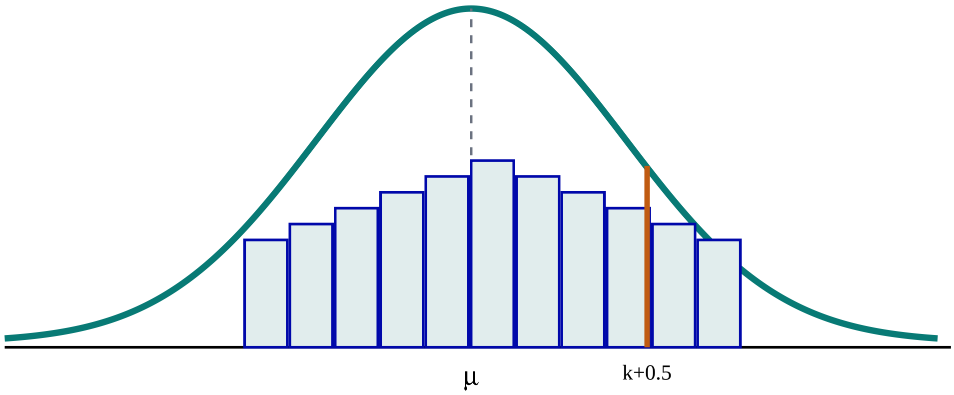
知识点 06/07 | 双边区间的连续性修正 | Correcting both ends of an interval

题目 | Question. 同一模型，近似求$P(75\leqslant X\leqslant 90)$。

正确答案 | Correct answer.
修正为$P(74.5<Y<90.5)$。$z_1\approx -0.794$，$z_2\approx 1.516$。概率$=\Phi(1.516)-\Phi(-0.794)\approx 0.722$。

知识点 07/07 | 条件不满足时不用近似 | Rejecting an unsuitable approximation

$P(X\geqslant 70)$使用69.5，不是70.5；“more than 70”是$X\geqslant 71$，修正同样为$Y>70.5$。若np或nq太小，正态形状可能很差。



连续性修正把某个整数对应的整根离散柱转换为连续区间的半单位边界。

逐题精讲丨Exam workshop（共 3 项）

精讲 01/03丨写近似分布并修正边界丨Distribution and continuity correction

题目丨Question. For $X \sim B(120,0.4)$, write the approximating normal distribution and approximate $P(X \leqslant 42)$.

完整解答丨Full solution. Mean=np=48; variance=np(1−p)=120(0.4)(0.6)=28.8. Let $Y \sim N(48,28.8)$. Continuity correction changes $X \leqslant 42$ to $Y<42.5$. $z=(42.5-48)/\sqrt{28.8} \approx -1.025$, so the probability is approximately $\Phi(-1.025) \approx 0.153$.

精讲 02/03丨订正大于等于的边界丨Correcting a lower inclusive boundary

题目丨Question. A student changes $P(X \geqslant 70)$ to $P(Y \geqslant 70.5)$. Correct the boundary and explain.

完整解答丨Full solution. $X \geqslant 70$ includes the entire bar centred at 70, whose lower edge is 69.5. The corrected event is therefore $P(Y>69.5)$. Using 70.5 would omit the probability corresponding to $X=70$.

精讲 03/03丨检验近似条件丨Checking suitability

题目丨Question. For $X \sim B(50,0.04)$, evaluate whether a normal approximation is suitable.

参考答案丨Model answer. np=50(0.04)=2 and n(1−p)=48. The first expected count is too small for a balanced bell shape, so a normal approximation is likely poor. An exact binomial calculation is preferable.